PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Effectiveness of Virtual Reality-Based Interventions in Rehabilitation Management of Breast Cancer Survivors: Protocol of a Systematic Review and Meta-Analysis
AUTHORS	Bu, Xiaofan; Ng, Peter H.F.; Chen, Qi; Cheng, Andy S. K.; Xu, Wenjing; Tong, Ying; Tang, Qingping; Liu, Xiangyu

VERSION 1 – REVIEW

REVIEWER	Tibau, Ariadna Sant Pau and Universitat Autònoma de Barcelona
REVIEW RETURNED	02-Aug-2021

GENERAL COMMENTS	The authors provide a valuable service to the oncology community by planning quantitatively evaluate the efficacy of virtual reality-based intervention in breast cancer-related rehabilitation management. The study question is relevant and the effectiveness of this intervention not convincingly demonstrated. Having said that, a few specific comments follow: 1. Although primary objective is clearly stated, I suggest to also report secondary objectives of the meta-analysis with pre-defined eligibility criteria. I suggest complete this part with explicit and reproducible methodology given the probably heterogeneity between studies. A data abstraction form will be also helpful. 2. I suggest to also include inclusion and exclusion criteria for the studies included in the meta-analysis. 3. Although the primary objective of the meta-analysis is to determine the effectiveness of VR-based intervention in breast cancer-related rehabilitation management I suggest also authors consider how the information extracted from this data will help to determine the appropriate sample size for a future trial, will help to highlight the weaknesses of previous studies and to recommend how to improve the design of future RCT trials. 4. Which period will be covered? 5. Could the authors provide in the protocol the keywords used for their research?

REVIEWER	Rynar, Lauren Rush University, Psychology
REVIEW RETURNED	25-Aug-2021

GENERAL COMMENTS	Review: Efficacy of Virtual Reality-Based Interventions in Rehabilitation Management of Breast Cancer Survivors: Protocol of a systematic review and meta-analysis

	Overall, I believe this will make for an interesting systematic review and meta-analysis on a relevant topic. It will be beneficial for researchers and clinicians interested in implementing virtual reality to understand the potential utility for managing cancer-related distress among breast cancer survivors. However, the current iteration needs revision for publication, largely due to wording, grammar, and typos, making the manuscript somewhat difficult to read. Specific suggestions are outlined below: 1. Abstract a. Introduction i. Second to last sentence: connect to previous sentence. “Virtual reality (VR) technology may offer new and noninvasive approaches to manage these adverse effects.” ii. Last sentence: Reword makes it seem you are presenting a VR protocol, rather than presenting a protocol to conduct a systematic review. b. Methods and analysis i. Clarify: is this a systematic review, meta-analysis, or both? At times sounds like you are interchanging, other times sounds like you are doing both. Some places you mention only one. ii. Again reword: This protocol for conducting a systematic review/meta-analysis was prepared according to… c. Ethics and dissemination i. Several typos/missing words. Review. ii. You only mention systematic review, not meta-analysis. 2. Strengths and limitations a. First bullet point: redundant. Delete after last comma. b. Last bullet point: redundant of first bullet point. 3. Introduction a. Several typos/missing words throughout. Review. b. Overall seems too much focus on cancer- and treatment-related adverse effects, and insufficient focus on VR. I might consider providing a brief review of cancer-related distress, and provide more background on what VR is, how it applies to cancer generally, and how it applies to rehabilitation after breast cancer. c. Define rehabilitation management. 4. Objectives: Written in past tense. This is a proposal. 5. Types of participations/populations/intervention: I suggest commenting on your broad inclusion criteria and how this will be accounted for in your review. 6. Data Collection/Analysis a. Suggest reorganizing to make clear how the systematic review will be performed vs the meta-analysis. Again, not always clear which is being referred to. 7. Data Synthesis: swap “analyze” with “analyses” 8. Discussion: a. Several typos/missing words throughout. Review. b. The second paragraph seems like it would fit better in the introduction than the discussion. Thank you for the opportunity to review this paper.

REVIEWER	Carvalho, Leonardo Cesar Universidade Federal de Alfenas
REVIEW RETURNED	15-Nov-2021

GENERAL COMMENTS	Dears,

	Although it is very well written and will surely be well received by readers, small considerations are necessary and have already been reported below: Overall: The present manuscript has its merit and reports on the protocol for a systematic review and meta-analysis exploring the efficacy of VR-based interventions on rehabilitation management of breast cancer survivors. Specific comments: Abstract It is adequate for the purpose of the study protocol. There are two databases that will be searched that are cited in PROSPERO and are not cited in the study. Keywords These should include words that are already in the title. Introduction It is adequate for the purpose of the study. Methods The methodology used in the protocol of a systematic review is adequate and was very well described. Discussion It is adequate for the purpose of the study. Tables It is adequate for the purpose of the study. References It is adequate for the purpose of the study. Best Regards

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Ariadna Tibau, Sant Pau and Universitat Autònoma de Barcelona

1Although primary objective is clearly stated, I suggest to also report secondary objectives of the meta-analysis with pre-defined eligibility criteria. I suggest complete this part with explicit and reproducible methodology given the probably heterogeneity between studies. A data abstraction form will be also helpful.
Response: Thank you. We have revised in Objectives section(page 5-6): “The overarching objective of this systematic review and meta-analysis is to report on studies using VR-based interventions for rehabilitation management in BCS and to quantitatively evaluate the efficacy of VR-based interventions in breast cancer-related rehabilitation management. This was achieved with four secondary objectives: (1) to identify the intervention type (immersive and non-immersive) that is most effective in breast cancer-related rehabilitation management; (2) to identify the most effective duration of intervention; and (3) to assess any lasting effect of VR-based interventions. (4) to use a simulation approach to estimate the sample size based on a meta-analysis of the existing evidence to recommend an appropriate sample size for a future trial.”

2 I suggest to also include inclusion and exclusion criteria for the studies included in the meta-analysis.
Response: Thank you. The inclusion and exclusion criteria (page 6-7) for the studies included in the meta-analysis are the same with systematic review. The outcomes that we extracted from the previous literature are all with respect to symptoms and problems that breast cancer survivors may be faced with. We will use qualitative and quantitative methods to evaluate the effectiveness of VR-based interventions in rehabilitation management of patients with breast cancer. If some outcomes could not be analyzed quantitatively such as data format (e.g. incidence of complications, cybersickness symptoms) or the fact that the data were only reported in one study (e.g., fatigue, fear of movement). In addition, we added exclusion criteria in (page 7):"Exclusion criteria: The following will be excluded: (1) did not specify the type of cancer; (2) described the technologies only; and (3) were conference papers, workshop papers, literature reviews, posters, comments, letters, study protocols, or proceedings papers.."

3Although the primary objective of the meta-analysis is to determine the effectiveness of VR-based intervention in breast cancer-related rehabilitation management I suggest also authors consider how the information extracted from this data will help to determine the appropriate sample size for a future trial, will help to highlight the weaknesses of previous studies and to recommend how to improve the design of future RCT trials.
Response: Thank you. We have added in the Objective section(page 6): "We will use simulation approach[1] to estimate the sample size based on meta-analysis of the existing evidence to recommend appropriate sample size for a future trial."

4Which period will be covered?
Response: Thank you. During or after breast cancer treatment.

5 Could the authors provide in the protocol the keywords used for their research?
Response: Thank you. The keywords section (page 3) reads: "Breast cancer survivors; virtual reality; rehabilitation management; systematic review; meta-analysis; effectiveness; protocol;"

Reference:
[1] Sutton AJ, Cooper NJ, Jones DR, et al. Evidence-based sample size calculations based upon updated meta-analysis. STAT MED 2007;26:2479-2500.

Reviewer: 2
Dr. Lauren Rynar, Rush University

1 Abstract
a Introduction
i Second to last sentence: connect to previous sentence. "Virtual reality (VR) technology may offer new and noninvasive approaches to manage these adverse effects."
Response: Thank you. We have revised in Introduction section (page 2): "Virtual reality (VR) technology uses computer technology to create an interactive three-dimensional world by visual, audio, and touch simulation being used in breast cancer rehabilitation management."

ii Last sentence: Reword makes it seem you are presenting a VR protocol, rather than presenting a protocol to conduct a systematic review.
Response: Thank you. We have deleted last sentence.

b Methods and analysis

i Clarify: is this a systematic review, meta-analysis, or both? At times sounds like you are interchanging, other times sounds like you are doing both. Some places you mention only one.
Response: Thank you. Both. I have revise it in Methods and analysis section (page 2): “the risk of bias for the studies included in the systematic review and meta-analysis will be assessed”. In addition, we checked the manuscript throughout.

ii Again reword: This protocol for conducting a systematic review/meta-analysis was prepared according to…
Response: Thank you. I have revised in Methods and analysis section (page 2): “This protocol for conducting a systematic review and meta-analysis was prepared according to the recommendations from Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement.”

c Ethics and dissemination
i Several typos/missing words. Review.
Response: Thank you. We apologize for our poor English. We have submitted the revised manuscript to a professional English editor to ensure the grammatical accuracy of the manuscript.

ii You only mention systematic review, not meta-analysis.
Response: Thank you. We have added “meta-analysis” in Ethics and dissemination section (page 2): “Ethics approval will not be needed because the data used in this systematic review and meta-analysis will be extracted from published studies. “

2 Strengths and limitations
a First bullet point: redundant. Delete after last comma.
Response: Thank you. We have deleted (page 3).

b Last bullet point: redundant of first bullet point.
Response: Thank you. We have deleted the last bullet point.

3 Introduction
a Several typos/missing words throughout. Review.
Response: Thank you. We apologize for our poor English. We have submitted the revised manuscript to a professional English editor to ensure the grammatical accuracy of the manuscript.

b Overall seems too much focus on cancer- and treatment-related adverse effects, and insufficient focus on VR. I might consider providing a brief review of cancer-related distress, and provide more background on what VR is, how it applies to cancer generally, and how it applies to rehabilitation after breast cancer.
Response: Thank you. We have deleted many contents regarding cancer- and treatment-related adverse effects, and provide more background on what VR is, how it applies to cancer generally, and how it applies to rehabilitation after breast cancer(page 4-5): “Virtual reality (VR) technology uses computer technology to create an interactive three-dimensional world by visual, audio, and touch simulation being used in breast cancer rehabilitation management. The development of VR offers new and noninvasive approaches for different healthcare applications. It creates interactive computer-generated worlds through visual, listening, and touch simulations, where an individual feels being. VR as a distraction tool improves cancer patients’ emotional well-being and decreases cancer-related psychological symptoms, enhancing their quality of life. In addition, VR-based rehabilitation systems (e.g., Xbox 360 Kinect games, the BrightArm Duo Rehabilitation System) have been identified to be effective for patients with weak arms and diminished grasping ability. These systems has been determined to be effective for patients with weak arms and diminished ability to grasp. It uses VR to

engage the patients in upper body bimanual exercises and simultaneously provides cognitive training and affective relief."

c Define rehabilitation management.
Response: Thank you. Rehabilitation management can refer to how to coordinate or manage a multidisciplinary rehabilitation team[2].

4 Objectives: Written in past tense. This is a proposal.
Response: Thank you. We have revised..

5 Types of participations/populations/intervention: I suggest commenting on your broad inclusion criteria and how this will be accounted for in your review.
Response: Thank you.
Types of participants/population
To obtain comprehensive studies and reduce selective bias, so: "we will include all trials that have included patients after first or recurrent breast cancer, regardless of demographic age, race, and education status."(page 6)

Types of interventions
To obtain comprehensive studies and reduce selective bias, we will select all trials assessing VR-based interventions, regardless immersive and non-immersive virtual environment: "We will select all trials assessing VR-based interventions (immersive and non-immersive virtual environment) regardless of their aim. In trials, the selection process will not be based on the type or the nature of the VR-based interventions (immersive and non-immersive virtual environments)."(page 6)

6 Data Collection/Analysis
a Suggest reorganizing to make clear how the systematic review will be performed vs the meta-analysis. Again, not always clear which is being referred to.
Response: Thank you. We have revised in page 9-10. We will use qualitative and quantitative methods to evaluate the effectiveness of VR-based interventions in rehabilitation management of patients with breast cancer. If some outcomes could not be analyzed quantitatively such as data format or some data were only reported in one study. We searched the available literature with regard to VR-based interventions applied to breast cancer survivors. Outcomes with respect to symptoms and problems that breast cancer survivors may be faced with will be extracted from the previous literature.

7Data Synthesis: swap "analyze" with "analyses"
Response: Thank you. We have swapped "analyze" with "analyses"

8Discussion:
a Several typos/missing words throughout. Review.
Response: Thank you. We apologize for our poor English. We have submitted the revised manuscript to a professional English editor to ensure the grammatical accuracy of the manuscript.

b The second paragraph seems like it would fit better in the introduction than the discussion.
Response: Thank you. We have moved the second paragraph in Discussion section to the Introduction section.

Reference:
[2]Case LE, Apkon SD, Eagle M, et al. Rehabilitation Management of the Patient With Duchenne Muscular Dystrophy. PEDIATRICS 2018;1422:S17-S33.

Reviewer: 3
Dr. Leonardo Cesar Carvalho, Universidade Federal de Alfenas
Comments to the Author:

Specific comments:
Abstract
It is adequate for the purpose of the study protocol. There are two databases that will be searched that are cited in PROSPERO and are not cited in the study.
Response: Thank you. We are sorry for our unrigorous writing. CINAHL belongs to EBSCO, and most journals associated with nursing and comprehensive healthcare are in the CINAHL database. Therefore, we searched CINAHL instead of EBSCO. Unfortunately, we do not have access to the Ovid database. Thus we did not include it in our search. Additionally, we considered the Cochrane Central Register of Controlled Trials essential and required. Hence we decided to add this database.

Keywords
These should include words that are already in the title.
Response: Thank you. We added “effectiveness; protocol” in keywords, so keywords section now reads “Breast cancer survivors; virtual reality; rehabilitation management; systematic review; meta-analysis; effectiveness; protocol;”

Introduction
It is adequate for the purpose of the study.
Response: Thank you.

Methods
The methodology used in the protocol of a systematic review is adequate and was very well described.
Response: Thank you.

Discussion
It is adequate for the purpose of the study.
Response: Thank you.

Tables
It is adequate for the purpose of the study.
Response: Thank you.

References
It is adequate for the purpose of the study.
Response: Thank you.

VERSION 2 – REVIEW

REVIEWER	Tibau, Ariadna Sant Pau and Universitat Autònoma de Barcelona
REVIEW RETURNED	21-Dec-2021

GENERAL COMMENTS	I have no further comments. Congratulations to the authors for their review. Thanks.